

## Analytic Geometry — Practice Problem Set

**Q1.** (4 marks)

On  $\mathbb{R}^3$  with the standard (Euclidean) inner product, let  $u = (1, 1, 0)^\top$  and  $v = (0, 1, 1)^\top$ .

- (a) Find *all* vectors  $w \in \mathbb{R}^3$  orthogonal to *both*  $u$  and  $v$ , and write down the *unit* vectors of this form. [2 M]
- (b) Compute the angle  $\omega$  between  $u$  and  $v$ . [1 M]
- (c) Show that  $\{u, v, w\}$  is a basis of  $\mathbb{R}^3$ , and explain geometrically why  $w$  is perpendicular to the plane spanned by  $u$  and  $v$ . [1 M]

### Solution

(a) *Idea:* “orthogonal to both” means two dot-product equations. Write  $w = (w_1, w_2, w_3)^\top$ :

$$w^\top u = w_1 + w_2 = 0, \quad w^\top v = w_2 + w_3 = 0.$$

From these,  $w_1 = -w_2$  and  $w_3 = -w_2$ . Letting the free variable be  $w_2 = t$ ,

$$w = t(-1, 1, -1)^\top, \quad t \in \mathbb{R}.$$

To get *unit* vectors, divide by the length  $\|(-1, 1, -1)^\top\| = \sqrt{3}$ :

$$w = \pm \frac{1}{\sqrt{3}}(-1, 1, -1)^\top.$$

(b) *Use*  $\cos \omega = \frac{u^\top v}{\|u\| \|v\|}$ . Here  $u^\top v = 0 + 1 + 0 = 1$  and  $\|u\| = \|v\| = \sqrt{2}$ , so

$$\cos \omega = \frac{1}{\sqrt{2} \cdot \sqrt{2}} = \frac{1}{2} \implies \boxed{\omega = 60^\circ}.$$

(c) *Basis test:* three vectors in  $\mathbb{R}^3$  form a basis exactly when their determinant is nonzero. With  $w = (-1, 1, -1)^\top$ ,

$$\det[u \ v \ w] = \begin{vmatrix} 1 & 0 & -1 \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{vmatrix} = -3 \neq 0,$$

so  $\{u, v, w\}$  is a basis. Geometrically,  $u$  and  $v$  span a plane; since  $w$  was constructed to be orthogonal to both  $u$  and  $v$ , it is orthogonal to *every* vector in that plane, i.e.  $w$  is the normal direction to the plane.

**Q2.** (4 marks)

(*Similarity vs. dissimilarity of data objects.*) Three data points live in  $\mathbb{R}^3$ :

$$p_1 = (2, 1, 0)^\top, \quad p_2 = (0, 1, 2)^\top, \quad p_3 = (2, 1, 1)^\top.$$

- (a) Build the  $3 \times 3$  dissimilarity matrix  $D$  using Euclidean distance. [2 M]

- (b) Which two points are the *most similar*? [1 M]
- (c) Compute the cosine similarity between  $p_1$  and  $p_3$ . [1 M]

### Solution

(a) *Dissimilarity* between two points is just the distance  $d(p_i, p_j) = \|p_i - p_j\|$ . Compute each pair:

$$\begin{aligned} d(p_1, p_2) &= \|(2, 0, -2)^\top\| = \sqrt{8} = 2\sqrt{2}, & d(p_1, p_3) &= \|(0, 0, -1)^\top\| = 1, \\ d(p_2, p_3) &= \|(-2, 0, 1)^\top\| = \sqrt{5}. \end{aligned}$$

Place these in a symmetric matrix with zeros on the diagonal (a point's distance to itself is 0):

$$D = \begin{bmatrix} 0 & 2\sqrt{2} & 1 \\ 2\sqrt{2} & 0 & \sqrt{5} \\ 1 & \sqrt{5} & 0 \end{bmatrix}.$$

(b) “Most similar” = *smallest* dissimilarity. The smallest off-diagonal entry is  $d(p_1, p_3) = 1$ , so  $p_1$  and  $p_3$  are the most similar.

(c) *Cosine similarity* is  $\cos \theta = \frac{p_1^\top p_3}{\|p_1\| \|p_3\|}$ :

$$p_1^\top p_3 = 2 \cdot 2 + 1 \cdot 1 + 0 \cdot 1 = 5, \quad \|p_1\| = \sqrt{5}, \quad \|p_3\| = \sqrt{6},$$

$$\cos \theta = \frac{5}{\sqrt{5}\sqrt{6}} = \frac{5}{\sqrt{30}} = \frac{\sqrt{30}}{6} \approx 0.913.$$

(Close to 1, confirming  $p_1$  and  $p_3$  point in nearly the same direction.)

### Q3.

(4 marks)

Let  $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$  and define a new inner product on  $\mathbb{R}^2$  by  $\langle x, y \rangle_A = x^\top A y$ .

- (a) Show  $A$  is symmetric positive definite, so  $\langle \cdot, \cdot \rangle_A$  is a genuine inner product. [1 M]
- (b) The vectors  $x = (1, 0)^\top$  and  $y = (0, 1)^\top$  are orthogonal in the *dot* product. Are they still orthogonal under  $\langle \cdot, \cdot \rangle_A$ ? Find the angle between them in this new inner product. [2 M]
- (c) Find a vector  $y' = (a, 1)^\top$  that is orthogonal to  $x$  with respect to  $\langle \cdot, \cdot \rangle_A$ . [1 M]

### Solution

(a)  $A$  is symmetric since  $A^\top = A$ . *Sylvester's criterion* (all leading minors positive):

$$D_1 = 2 > 0, \quad D_2 = \det A = 2 \cdot 2 - 1 \cdot 1 = 3 > 0.$$

So  $A$  is positive definite. A symmetric, positive-definite matrix always makes  $x^\top A y$  a valid inner product (it is bilinear, symmetric, and  $x^\top A x > 0$  for  $x \neq 0$ ).

(b) *Test orthogonality* by computing  $\langle x, y \rangle_A = x^\top Ay$ . First  $Ay = A(0, 1)^\top = (1, 2)^\top$ , hence

$$\langle x, y \rangle_A = x^\top (Ay) = (1, 0) \cdot (1, 2) = 1 \neq 0.$$

They are **not** orthogonal here. To get the angle, we also need the  $A$ -norms:

$$\|x\|_A = \sqrt{x^\top Ax} = \sqrt{A_{11}} = \sqrt{2}, \quad \|y\|_A = \sqrt{A_{22}} = \sqrt{2}.$$

Therefore

$$\cos \theta = \frac{\langle x, y \rangle_A}{\|x\|_A \|y\|_A} = \frac{1}{\sqrt{2} \cdot \sqrt{2}} = \frac{1}{2} \implies \theta = 60^\circ.$$

*Takeaway:* orthogonality depends on *which* inner product you use.

(c) We need  $\langle x, y' \rangle_A = 0$ . Note  $x^\top A = (1, 0)A = (2, 1)$ , so

$$\langle x, y' \rangle_A = (2, 1) \cdot (a, 1) = 2a + 1 = 0 \implies a = -\frac{1}{2}.$$

Thus  $y' = (-\frac{1}{2}, 1)^\top$ .

**Q4.**

**(5 marks)**

On  $\mathbb{R}^3$  use the weighted inner product  $\langle x, y \rangle_A = x^\top Ay$  with  $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$  (SPD). Let

$a = (1, -1, 0)^\top$  and  $b = (1, 0, 2)^\top$ .

(a) Compute  $\langle b, a \rangle_A$ ,  $\|a\|_A$ , and  $\|b\|_A$ . [2 M]

(b) Find the orthogonal projection  $p$  of  $b$  onto the line spanned by  $a$ , *in this inner product*. [2 M]

(c) Verify that the residual  $b - p$  is  $A$ -orthogonal to  $a$  (this is what makes  $p$  the *orthogonal* projection). [1 M]

### Solution

(a) *Trick:* compute the matrix-vector product once, then take dot products. With  $Aa = A(1, -1, 0)^\top = (1, -1, 0)^\top$ :

$$\langle b, a \rangle_A = b^\top (Aa) = (1, 0, 2) \cdot (1, -1, 0) = 1, \quad \|a\|_A^2 = a^\top (Aa) = (1, -1, 0) \cdot (1, -1, 0) = 2.$$

For  $\|b\|_A$ , use  $Ab = A(1, 0, 2)^\top = (2, 1, 2)^\top$ :

$$\|b\|_A^2 = b^\top (Ab) = (1, 0, 2) \cdot (2, 1, 2) = 6.$$

So  $\langle b, a \rangle_A = 1$ ,  $\|a\|_A = \sqrt{2}$ ,  $\|b\|_A = \sqrt{6}$ .

(b) *Projection formula* (same as usual, but every inner product is the  $A$ -one):

$$p = \frac{\langle b, a \rangle_A}{\langle a, a \rangle_A} a = \frac{1}{2} (1, -1, 0)^\top = \left(\frac{1}{2}, -\frac{1}{2}, 0\right)^\top.$$

(c) The residual is

$$r = b - p = (1, 0, 2)^\top - \left(\frac{1}{2}, -\frac{1}{2}, 0\right)^\top = \left(\frac{1}{2}, \frac{1}{2}, 2\right)^\top.$$

*Check A-orthogonality* using the same  $Aa = (1, -1, 0)^\top$ :

$$\langle r, a \rangle_A = r^\top (Aa) = \left(\frac{1}{2}, \frac{1}{2}, 2\right) \cdot (1, -1, 0) = \frac{1}{2} - \frac{1}{2} + 0 = 0. \checkmark$$

Since  $r \perp_A a$ ,  $p$  is indeed the orthogonal projection of  $b$  onto the line through  $a$ .

**Q5.**

**(5 marks)**

Consider the symmetric matrix  $A(t) = \begin{bmatrix} t & 3 \\ 3 & t \end{bmatrix}$ ,  $t \in \mathbb{R}$ .

- (a) Classify  $A(t)$  for *all* real  $t$  as positive definite, positive semidefinite, indefinite, negative semidefinite, or negative definite. [3 M]
- (b) For which  $t$  does  $\langle x, y \rangle = x^\top A(t) y$  define a valid inner product? At the borderline value  $t = 3$ , exhibit a nonzero vector  $v$  with  $\langle v, v \rangle = 0$ . [2 M]

### Solution

(a) *Method:* a symmetric matrix is classified by the signs of its eigenvalues. For  $A(t) = \begin{bmatrix} t & 3 \\ 3 & t \end{bmatrix}$ , solve  $\det(A - \lambda I) = 0$ :

$$(t - \lambda)^2 - 9 = 0 \implies t - \lambda = \pm 3 \implies \lambda = t - 3 \text{ or } \lambda = t + 3.$$

Now just read off the signs of the two eigenvalues  $t - 3$  and  $t + 3$ :

$$\begin{array}{ll} t > 3 : & \text{both positive} \Rightarrow \textbf{positive definite} \\ t = 3 : & \lambda = 6, 0 \Rightarrow \textbf{positive semidefinite} \\ -3 < t < 3 : & \text{one+}, \text{one-} \Rightarrow \textbf{indefinite} \\ t = -3 : & \lambda = 0, -6 \Rightarrow \textbf{negative semidefinite} \\ t < -3 : & \text{both negative} \Rightarrow \textbf{negative definite} \end{array}$$

(b) An inner product needs a *positive-definite* matrix, so we need  $t > 3$ . At  $t = 3$  the matrix is only *semidefinite*, which fails the rule  $\langle v, v \rangle > 0$  for  $v \neq 0$ . Concretely, take  $v = (1, -1)^\top$ .

With  $A(3) = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}$ ,

$$\langle v, v \rangle = v^\top A(3) v = (1, -1) \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = (1, -1) \cdot (0, 0) = 0,$$

even though  $v \neq 0$ . So  $t = 3$  does not give an inner product.

**Q6.**

**(5 marks)**

(Positive definiteness over  $\mathbb{C}$ .) For a *real* parameter  $k$ , consider

$$A(k) = \begin{bmatrix} k & 1+i \\ 1-i & 2 \end{bmatrix}.$$

For  $u, v \in \mathbb{C}^2$ , let  $u^*$  denote the conjugate transpose and set  $\langle u, v \rangle = u^* A(k) v$ .

- (a) Show  $A(k)$  is Hermitian for every real  $k$ . [1 M]
- (b) Find all real  $k$  for which  $A(k)$  is positive definite — equivalently, for which  $\langle u, v \rangle = u^* A(k) v$  is a valid inner product on  $\mathbb{C}^2$ . [2 M]
- (c) At the borderline value  $k = 1$  the matrix is only positive *semidefinite*. Find a nonzero vector  $z \in \mathbb{C}^2$  with  $z^* A(1) z = 0$ , showing the form degenerates there. [2 M]

### Solution

(a) A matrix is *Hermitian* when  $A^* = A$ , i.e. the diagonal entries are real and the two off-diagonal entries are complex conjugates of each other. Here the diagonal is  $k, 2$  (real, because  $k$  is real) and the off-diagonals satisfy  $\overline{1+i} = 1-i$ . Hence  $A(k)^* = A(k)$  for every real  $k$ .

(b) *Sylvester's criterion* also works for Hermitian matrices: positive definite  $\iff$  all leading principal minors are positive (they come out real automatically). The minors are

$$D_1 = k, \quad D_2 = \det A(k) = 2k - (1+i)(1-i).$$

The crucial step is the complex product, which is a *squared modulus*:

$$(1+i)(1-i) = 1 - i^2 = 2 \quad (= |1+i|^2), \quad \text{so} \quad D_2 = 2k - 2 = 2(k-1).$$

Requiring  $D_1 > 0$  and  $D_2 > 0$  gives  $k > 0$  and  $k > 1$ , so

$$A(k) \text{ is positive definite } \iff k > 1.$$

(c) At  $k = 1$  we have  $D_2 = 0$ , so  $A(1) = \begin{bmatrix} 1 & 1+i \\ 1-i & 2 \end{bmatrix}$  is singular ( $D_1 = 1 > 0$ ,  $D_2 = 0$ : positive semidefinite). *Idea:* any nonzero null-space vector works, since  $A(1)z = 0 \implies z^* A(1) z = z^* \mathbf{0} = 0$ . From the first row,  $z_1 + (1+i)z_2 = 0$ ; choosing  $z_2 = 1$  gives  $z_1 = -(1+i)$ , so

$$z = \begin{pmatrix} -(1+i) \\ 1 \end{pmatrix}.$$

Verify it is a null vector (using  $(1-i)(1+i) = 2$ ):

$$A(1)z = \begin{pmatrix} 1 \cdot (-(1+i)) + (1+i) \cdot 1 \\ (1-i) \cdot (-(1+i)) + 2 \cdot 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -2+2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Hence  $z^* A(1) z = 0$  with  $z \neq 0$ : at  $k = 1$  the form is **not** positive definite, so it is not an inner product.

For each matrix  $M$  below, decide whether  $\langle x, y \rangle = x^\top M y$  defines a valid inner product on  $\mathbb{R}^2$ . Justify using the three requirements (bilinear, symmetric, positive-definite); when it fails, exhibit a witness.

$$(a) M = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \quad (b) M = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \quad (c) M = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

[2 M each]

### Solution

Every  $x^\top M y$  is automatically bilinear, so only *symmetry* and *positive-definiteness* can fail.

(a)  $M$  is symmetric, so symmetry holds. Check positive-definiteness by Sylvester:  $D_1 = 1 > 0$  but  $D_2 = \det M = 1 - 4 = -3 < 0$ . *Fails*. A witness:  $x = (1, -1)^\top$  gives

$$x^\top M x = (1, -1) \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = (1, -1) \cdot (-1, 1) = -2 < 0.$$

So  $\langle x, x \rangle < 0$  is possible — **not** an inner product.

(b)  $M$  is symmetric. Sylvester:  $D_1 = 2 > 0$  and  $D_2 = \det M = 6 - 1 = 5 > 0$ , so  $M$  is positive definite. All three requirements hold — this **is** a valid inner product.

(c)  $M$  is *not* symmetric, and that breaks the symmetry requirement directly. Test on the basis vectors:

$$\langle e_1, e_2 \rangle = e_1^\top M e_2 = M_{12} = 1, \quad \langle e_2, e_1 \rangle = e_2^\top M e_1 = M_{21} = 0.$$

Since  $\langle e_1, e_2 \rangle \neq \langle e_2, e_1 \rangle$ , symmetry fails — **not** an inner product (even though the quadratic form  $x^\top M x = x_1^2 + x_1 x_2 + x_2^2$  happens to be positive; symmetry alone already rules it out).

### Q8.

(5 marks)

On  $\mathbb{R}^2$  use  $\langle x, y \rangle_A = x^\top A y$  with  $A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$  (SPD), and let  $x = (1, 1)^\top$ ,  $y = (1, -1)^\top$ .

(a) Compute  $\|x\|_A$ ,  $\|y\|_A$ , and  $\langle x, y \rangle_A$ . [2 M]

(b) Verify the Cauchy–Schwarz inequality  $|\langle x, y \rangle_A| \leq \|x\|_A \|y\|_A$ . [1 M]

(c) Compute the distance  $d(x, y) = \|x - y\|_A$  and verify the triangle inequality  $\|x - y\|_A \leq \|x\|_A + \|y\|_A$ . [2 M]

### Solution

(a) *Compute  $Ax$  and  $Ay$  once.*  $Ax = A(1, 1)^\top = (4, 3)^\top$  and  $Ay = A(1, -1)^\top = (2, -1)^\top$ . Then

$$\|x\|_A^2 = x^\top (Ax) = (1, 1) \cdot (4, 3) = 7, \quad \|y\|_A^2 = y^\top (Ay) = (1, -1) \cdot (2, -1) = 3,$$

$$\langle x, y \rangle_A = x^\top (Ay) = (1, 1) \cdot (2, -1) = 1.$$

So  $\|x\|_A = \sqrt{7}$ ,  $\|y\|_A = \sqrt{3}$ ,  $\langle x, y \rangle_A = 1$ .

(b) *Cauchy–Schwarz:*

$$|\langle x, y \rangle_A| = 1 \leq \|x\|_A \|y\|_A = \sqrt{7} \cdot \sqrt{3} = \sqrt{21} \approx 4.58. \checkmark$$

(c) *Distance* uses the same norm on the difference. With  $x - y = (0, 2)^\top$  and  $A(0, 2)^\top = (2, 4)^\top$ :

$$\|x - y\|_A^2 = (0, 2) \cdot (2, 4) = 8 \implies d(x, y) = \sqrt{8} = 2\sqrt{2} \approx 2.83.$$

*Triangle inequality:*

$$\|x - y\|_A = 2\sqrt{2} \approx 2.83 \leq \|x\|_A + \|y\|_A = \sqrt{7} + \sqrt{3} \approx 4.38. \checkmark$$

*End of practice set — 8 questions, each with a full worked solution.*